Effects of social information on fricative discrimination in a change detection task

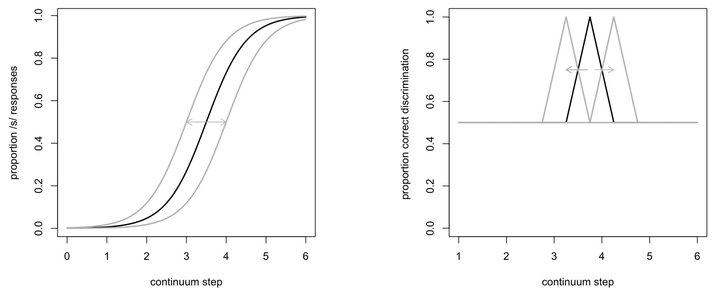
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4pSCa14

1. The Strand Effect in Identification

- The perception of speech is fundamentally multimodal (McGurk & MacDonald, 1976)
  - McGurk effect: co-occurrence of visual and auditory phonetic information causes fused perceptions (McGurk & MacDonald 1976; Munhall et al., 1996)
- Multimodal social bias speech sound identification
  - Strand effect: identical acoustics paired with gendered cues shift /s/-/ʃ/ boundary (May 1976; Strand & Johnson, 1996; Johnson et al., 1999)
- Unlike McGurk effect (e.g. Green et al., 1991), Strand effect disappears when expectations about gender of voice conflicts with gender of face (Laycock & McGowan, 2025)
  - In incongruent pairings, visual information about talkers seems to be unintegrated (e.g. Rubin, 1992; McGowan & Babel, 2020; Campbell-Kibler, 2021)
- What do listeners do with the faces in these cases?
  - One possibility: faces are still *perceived*, but the unexpected visual cues are inhibited in processing

Predicted boundary shifts in sibilant identification (left) and discrimination (right) due to male and female faces



Research Questions

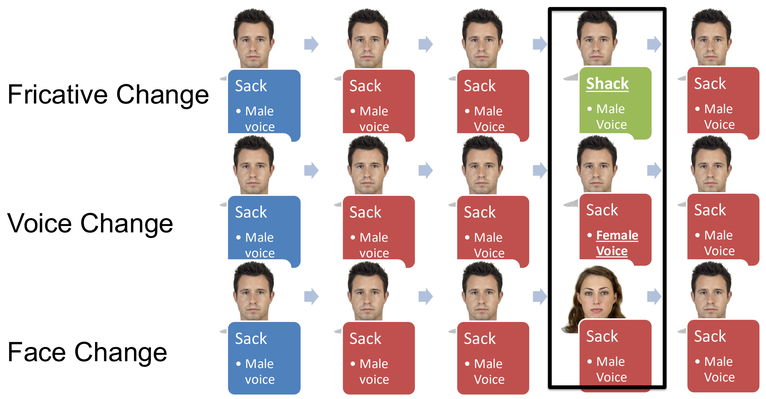
1. Can listeners inhibit the gendered face effect in sibilant discrimination as they seem to in identification?

2. If so, are there residual processing costs associated with doing so?

2. Change Detection Task

- Participants: N=18 (8 male, 10 female)
- Task Instructions: "press button when **word** changes"
- Materials:
  - 6-step *Sack-Shack* continuum (Bouavichith et al., 2019)
  - Male and female source voices (Bouavichith et al., 2019)
  - Male and female faces from Chicago Face Database (Ma et al., 2015)

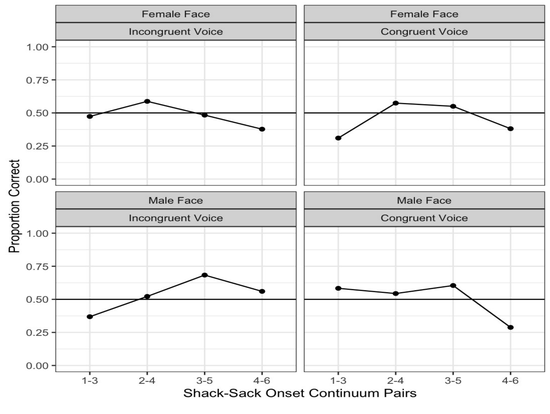
Trial Type	Tests:
Fricative Change	Strand effect in discrimination
Voice Change	Within gender: talker specificity; Across gender: attention to voice when it becomes incongruent
Face Change	Incongruent: if the effect persists Congruent: if the effect is restored



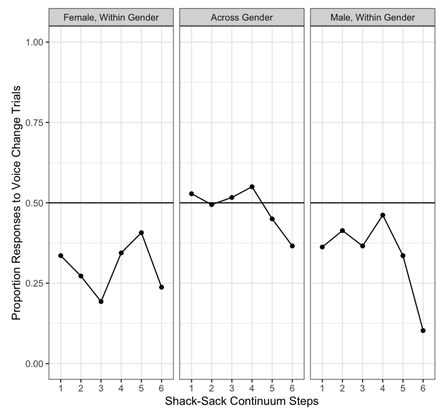
4. Preliminary Results

Trial Type	Percent Change Detection
Fricative Change	15%
Within Gender Voice Change	15%
Across Gender Voice Change	18%
Congruent Face Change	14%
Incongruent Face Change	8%

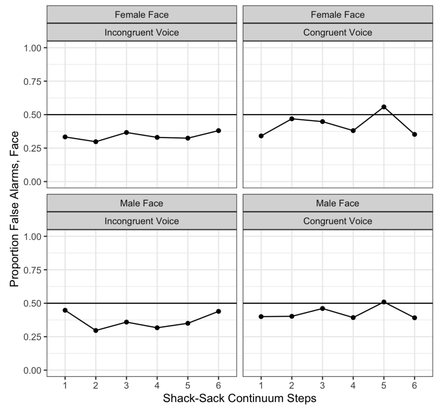
Fricative Change



Voice Change



Face Change



Trial Type	Mean Response Time
Fricative Change	237ms *(n=300)*
Within Gender Voice Change	361ms (*n=171)*
Across Gender Voice Change	286ms (*n=286)*
Congruent Face Change	402ms (*n=88)*
Incongruent Face Change	433ms *(n=71)*

5. Discussion

Fricative change:

- Expected discrimination pattern, generally
- Accuracy is lower than expected across boundaries (~60%)
- Strand effect is present
  - Male faces bias more of continuum toward /s/

False alarms not evenly distributed:

- Different magnitudes across trial types
- Comparable responses for voice change to accurate responses for fricative change
- Face change had fewer false alarms, especially when the change was incongruous

Listeners attend more to voice when modalities are incongruent

- Strand effect disappears when expectations about gender of voice conflicts with gender of face in discrimination
- Replicates findings from identification (Laycock & McGowan, 2025)
- But face false alarms are much slower, suggesting residual processing costs of this incongruity

Acknowledgements

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